Introduction

Probability Density Function

Cumulative Distribution Function

Function of RVs

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Expectation

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Common Distributions

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Continuous Random Variable and Probability Distributions

Hoàng Văn Hà
University of Science, VNU - HCM
hvha@hcmus.edu.vn

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Definition of Continuous Random Variable

Definition 1 (Continuous Random Variable)

A random variable X is called a continuous random variable *if its set of possible values is an interval or a union of intervals on the real line.*

Properties

- Can take *infinitely many values* in an interval
- Probability of taking any specific value is zero:

$$P(X = a) = 0 \quad \forall a \in \mathbb{R}.$$

- We can only compute probabilities over intervals:

$$P(a \leq X \leq b) > 0.$$

- Described by a **probability density function (PDF)**.

Discrete vs. Continuous Random Variables

Discrete RV

- Countable values
- $P(X = x_i) > 0$
- Probability mass function (PMF)
- Sum: $\sum P(X = x_i) = 1$

Examples:

- Number of heads in coin flips
- Number of customers
- Dice outcomes

Continuous RV

- Uncountable values
- $P(X = x) = 0$
- Probability density function (PDF)
- Integral: $\int_{-\infty}^{\infty} f(x)dx = 1$

Examples:

- Height, weight, temperature
- Time, distance
- Measurement errors

Examples of Continuous Random Variables

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Physical Measurements

- **Height of adults**: $X \in [100, 250]$ cm
- **Weight of newborns**: $X \in [1, 6]$ kg
- **Daily temperature**: $X \in [-50, 60]$ °C

Time-Related Variables

- **Waiting time at a bus stop**: $X \in [0, \infty)$ minutes
- **Lifespan of a light bulb**: $X \in [0, \infty)$ hours
- **Time to complete a task**: $X \in [0, T]$ minutes

Financial Variables

- **Stock price**: $X \in [0, \infty)$ dollars
- **Interest rate**: $X \in [0, 1]$ (proportion)
- **Investment return**: $X \in (-\infty, \infty)$ percent

Probability Density Function (PDF)

Definition

A function $f(x)$ is called a **probability density function (PDF)** of a continuous random variable X if:

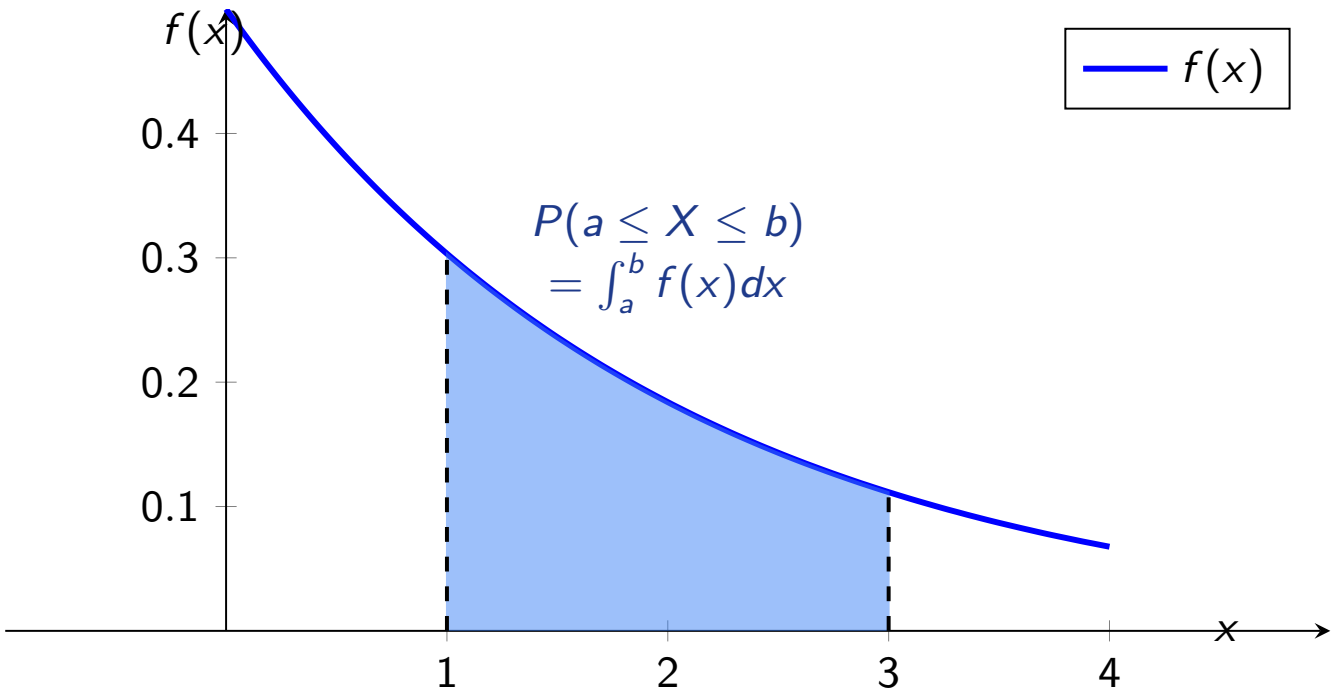
1. $f(x) \geq 0$ for all $x \in \mathbb{R}$.
2. $\displaystyle\int_{-\infty}^{\infty} f(x)dx = 1$.
3. For any interval $I \subset \mathbb{R}$:

$$P(X \in I) = \int_I f(x)dx.$$

Important Notes

- $f(x)$ is **NOT** a probability (it can be > 1).
- $f(x)$ represents **probability density**.
- $P(a \leq X \leq b) = P(a < X < b) = P(a \leq X < b) = P(a < X \leq b)$.
- The area under the curve from a to b gives the probability.

PDF of continuous random variables



- The **shaded area** represents $P(a \leq X \leq b)$.
- Total area under the curve $= 1$.

Example 1: Uniform Distribution

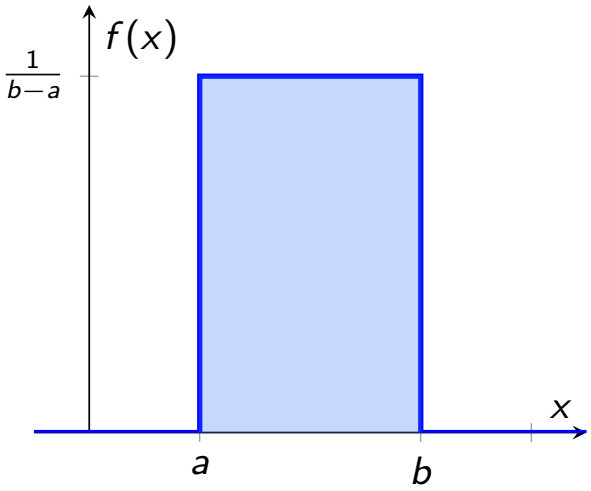
Uniform Distribution on $[a, b]$

A random variable X is uniformly distributed on $[a, b]$ if its PDF is:

$$f(x) = \begin{cases} \frac{1}{b-a} & \text{if } a \leq x \leq b \\ 0 & \text{otherwise.} \end{cases}$$

Properties:

- Equal probability density everywhere in $[a, b]$.
- $P(X \leq c) = \frac{c-a}{b-a}$ for $a \leq c \leq b$.



Example: Bus arrival time

- Buses arrive every 10 minutes;
- X = waiting time;
- $X \sim \text{Uniform}[0, 10]$;
- $f(x) = 0.1$ for $x \in [0, 10]$.

Example 2: Exponential Distribution

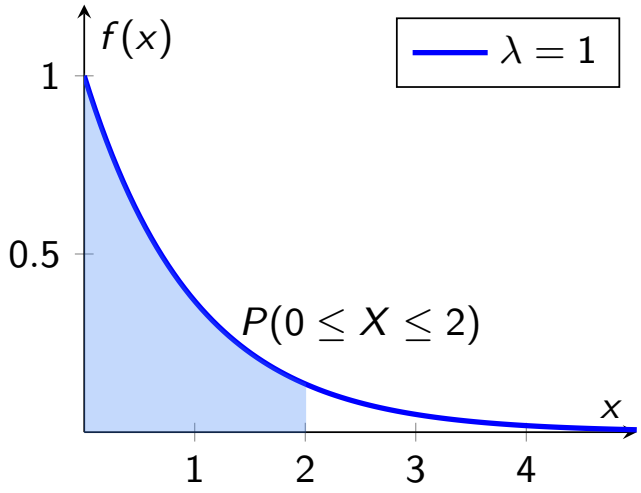
Exponential Distribution

A random variable X has an exponential distribution with parameter $\lambda > 0$ if:

$$f(x) = \begin{cases} \lambda e^{-\lambda x} & \text{if } x \geq 0 \\ 0 & \text{if } x < 0. \end{cases}$$

Properties:

- Models waiting times, lifespans.
- Mean: $E[X] = \frac{1}{\lambda}$.
- Memoryless property.



Example: Light bulb lifetime

- X = lifetime in hours;
- Average lifetime = 1000 hours;
- $\lambda = 0.001$;
- $f(x) = 0.001e^{-0.001x}$ for $x \geq 0$.

Cumulative Distribution Function (CDF)

Definition 2

The cumulative distribution function (CDF) *of a continuous random variable* X *is defined as:*

$$F(x) = P(X \leq x) = \int_{-\infty}^{x} f(t)dt.$$

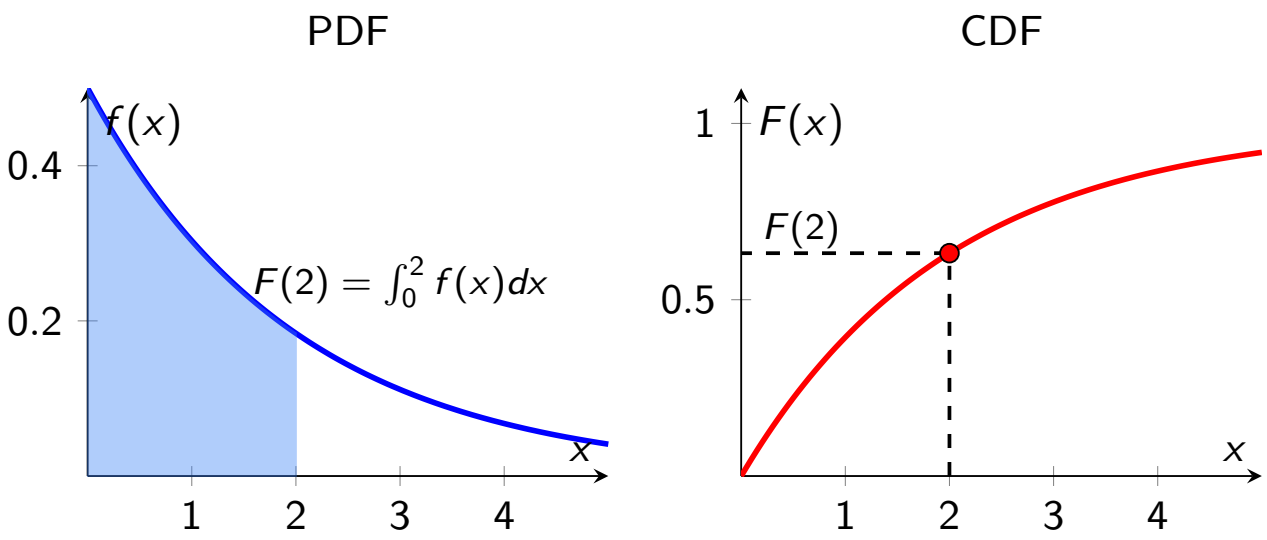
Properties of CDF

1. $0 \leq F(x) \leq 1$ for all x.
2. $F(x)$ is non-decreasing: if $x_1 < x_2$, then $F(x_1) \leq F(x_2)$.
3. $\lim_{x \to -\infty} F(x) = 0$ and $\lim_{x \to \infty} F(x) = 1$.
4. $F(x)$ is right-continuous.
5. $P(a < X \leq b) = F(b) - F(a)$.

Relationship between PDF and CDF

$$F'(x) = f(x) \quad \text{and} \quad f(x) = \frac{d}{dx}F(x).$$

Relationship between PDF and CDF



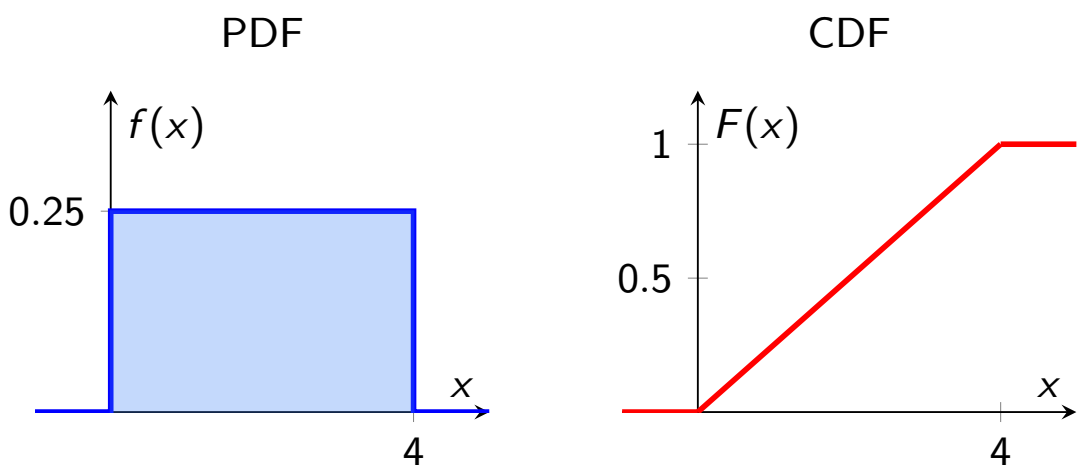
- The PDF $f(x)$ shows the **density** of probability.
- The CDF $F(x)$ shows the **accumulated** probability up to x.
- The area under PDF from $-\infty$ to x equals the CDF value at x.

Example 3: CDF of Uniform Distribution

Uniform Distribution on $[0, 4]$

PDF: $f(x) = \begin{cases} 0.25 & \text{if } 0 \leq x \leq 4 \\ 0 & \text{otherwise.} \end{cases}$

CDF: $F(x) = \begin{cases} 0 & \text{if } x < 0 \\ \frac{x}{4} & \text{if } 0 \leq x \leq 4 \\ 1 & \text{if } x > 4. \end{cases}$



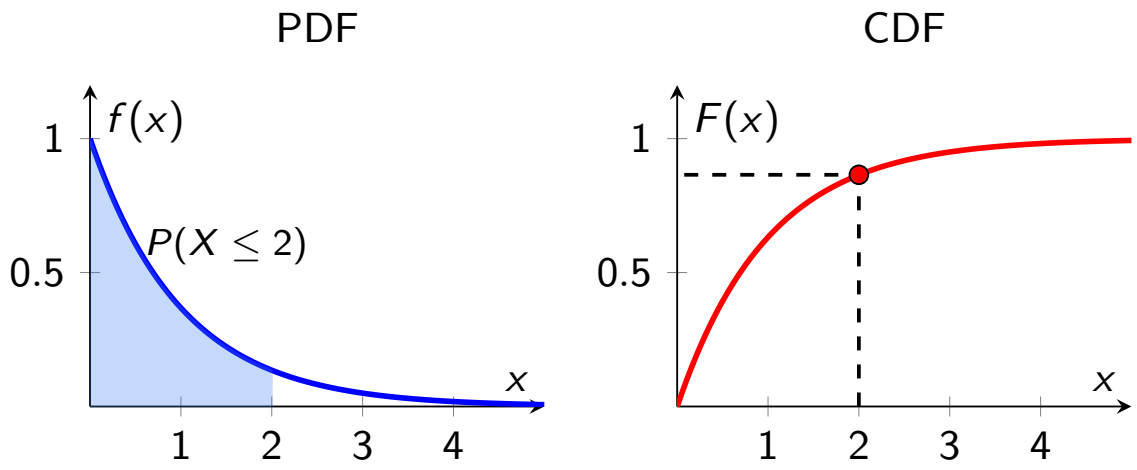
Calculation: $P(1 \leq X \leq 3) = F(3) - F(1) = \frac{3}{4} - \frac{1}{4} = 0.5.$

Example 4: CDF of Exponential Distribution

Exponential Distribution with $\lambda = 1$

PDF: $f(x) = \begin{cases} e^{-x} & \text{if } x \geq 0 \\ 0 & \text{if } x < 0. \end{cases}$

CDF: $F(x) = \begin{cases} 0 & \text{if } x < 0 \\ 1 - e^{-x} & \text{if } x \geq 0. \end{cases}$



Calculation: $P(X \leq 2) = F(2) = 1 - e^{-2} \approx 0.865.$

Example 5: Battery Lifetime

The lifetime X (in years) of a smartphone battery follows an exponential distribution with mean lifetime of 2 years.

a. Find the PDF and CDF of X.
b. Calculate $P(X > 3)$.
c. Calculate $P(1 < X < 4)$.
d. Find the median lifetime.

Introduction: Why study functions of RVs?

Motivation

Often, we observe a random variable X, but we're interested in some function of it: $Y = g(X)$.

Real-World Examples

- X = radius of a circle, $Y = \pi X^2$ = area;
- X = side length of a square, $Y = X^2$ = area;
- X = temperature in Celsius, $Y = \frac{9}{5}X + 32$ = temperature in Fahrenheit;
- X = investment amount, $Y = X(1 + r)$ = future value;
- X = raw test score, $Y = aX + b$ = standardized score;

Question

Given the pdf/cdf of X, how do we find the pdf/cdf of $Y = g(X)$?

Function of RVs

Given

- Random variable X with known PDF $f_X(x)$ and CDF $F_X(x)$.
- A function $g : \mathbb{R} \to \mathbb{R}$.
- Define $Y = g(X)$.

Goal

Find the distribution of Y:

- PDF: $f_Y(y)$.
- CDF: $F_Y(y) = P(Y \leq y) = P(g(X) \leq y)$.

$$X \sim f_X(x) \longrightarrow Y = g(X) \longrightarrow Y \sim f_Y(y)\ ?$$

Method 1: CDF Method (Universal Approach)

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The CDF Method

This method works for **any** function $g(X)$:

Step 1: Find the CDF of Y

$$F_Y(y) = P(Y \leq y) = P(g(X) \leq y).$$

Step 2: Express in terms of X

$$F_Y(y) = P(X \in A_y)$$

where $A_y = \{x : g(x) \leq y\}$.
Step 3: Compute using F_X or integrate f_X

$$F_Y(y) = \int_{A_y} f_X(x)dx.$$

Step 4: Differentiate to get PDF

$$f_Y(y) = \frac{d}{dy} F_Y(y).$$

Examples

Example 6

Let $X \sim \text{Uniform}(0, 1)$ with PDF: $f_X(x) = 1$ for $0 \leq x \leq 1$.
Find the PDF of $Y = 2X + 3$.

Example 7

Let $X \sim \text{Uniform}(-1, 1)$ with PDF: $f_X(x) = \dfrac{1}{2}$ for $-1 \leq x \leq 1$.

Find the PDF of $Y = X^2$.

Method 2: Change of variables formula

Theorem 1 (Transformation of variables for monotone functions)

Let X be the random variable with pdf $f_X(x)$. Let g be a one-to-one (injective) application and differentiable. Define $Y = g(X)$. If g is strictly monotone *(either always increasing or always decreasing) with inverse g^{-1}, then*

$$f_Y(y) = f_X(g^{-1}(y)) \cdot \left| \frac{d}{dy} g^{-1}(y) \right|.$$

or equivalently

$$f_Y(y) = f_X(x) \cdot \left| \frac{dx}{dy} \right| \quad \text{where } x = g^{-1}(y).$$

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Example 7

Let $X \sim \text{Exponential}(\lambda)$ with PDF: $f_X(x) = \lambda e^{-\lambda x}$ for $x > 0$.
Find the PDF of $Y = \sqrt{X}$.

Solution

Step 1: Find the inverse transformation

$$y = \sqrt{x} \quad \Rightarrow \quad x = y^2 = g^{-1}(y).$$

Note: g is strictly increasing on $(0, \infty)$.

Step 2: Compute

$$\frac{dx}{dy} = \frac{d}{dy}(y^2) = 2y.$$

Step 3: Apply the transformation formula

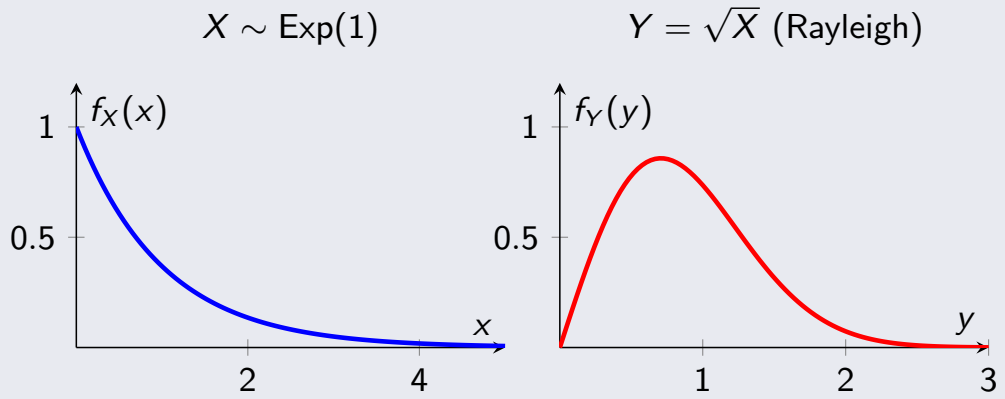
$$\begin{aligned} f_Y(y) &= f_X(y^2) \cdot |2y| = \lambda e^{-\lambda y^2} \cdot 2y \\ &= 2\lambda y e^{-\lambda y^2} \quad \text{for } y > 0. \end{aligned}$$

Example 7 (cont.)

We obtain:

$$f_Y(y) = \begin{cases} 2\lambda y e^{-\lambda y^2} & \text{if } y > 0 \\ 0 & \text{otherwise.} \end{cases}$$

This is a **Rayleigh distribution** with parameter λ.



Example 8: Exponential Transformation

Let $X \sim \text{Uniform}[0, 1]$. Find the PDF of $Y = -\ln(X)$.

Handling Non-Monotone Functions

Challenge

If g is **not** monotone (e.g., $Y = X^2$ where X can be negative), we need a different approach.

Solution: Partition the Domain

Step 1: Partition the domain of X into regions where g is monotone
Step 2: Apply the transformation formula on each region
Step 3: Sum the contributions:

$$f_Y(y) = \sum_i f_X(x_i) \cdot \left| \frac{dx_i}{dy} \right|$$

where x_i are all solutions to $g(x_i) = y$.

Handling Non-Monotone Functions - Examples

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Example 9

Let $X \sim \mathcal{N}(0,1)$ with PDF:

$$f_X(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}}, -\infty < x < \infty.$$

Find the PDF of $Y = X^2$.

Solution:
For $y > 0$, equation $x^2 = y$ has two solutions: $x_1 = \sqrt{y}$ and $x_2 = -\sqrt{y}$.

$$f_Y(y) = f_X(\sqrt{y}) \cdot \frac{1}{2\sqrt{y}} + f_X(-\sqrt{y}) \cdot \frac{1}{2\sqrt{y}}.$$

This gives the **chi-square distribution** with 1 degree of freedom.

Example 10

Let $X \sim \text{Uniform}[-1,1]$ with PDF: $f_X(x) = \frac{1}{2}$ for $-1 \leq x \leq 1$.
Find the PDF of $Y = |X|$.

Expected Value (Mean)

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Definition 3

The expected value *or* mean *of a continuous random variable X with pdf $f(x)$ is:*

$$\mu = E[X] = \int_{-\infty}^{\infty} x f(x) dx$$

provided the integral exists and is finite.

- The expected value represents the **long-run average** value.
- It is the **center of mass** or **balance point** of the distribution.
- Also called the **first moment** about the origin.
- Denoted as $E[X]$, μ, or μ_X.

Remark

The expected value may not be a value that the random variable can actually take! (e.g., $E[X] = 3.5$ for a die roll).

Properties of expected Value

Proposition 1 (Linearity of Expectation)

For constants a and b, and random variable X:

1. $E[c] = c$ *for any constant* c;
2. $E[aX] = aE[X]$, *this implies* $E[aX + b] = aE[X] + b$;
3. $E[X + Y] = E[X] + E[Y]$ *(even if X and Y are dependent).*

Example: Linear Transformation

If temperature in Celsius is C with $E[C] = 20$, then temperature in Fahrenheit $F = \frac{9}{5}C + 32$ has:

$$E[F] = \frac{9}{5}E[C] + 32 = \frac{9}{5}(20) + 32 = 36 + 32 = 68 \ (^{o}\text{F}).$$

Expected value of a function of X

If $g(X)$ is a function of X, then

$$E[g(X)] = \int_{-\infty}^{\infty} g(x)f(x)dx.$$

Important Special Cases

- **k-th moment about origin**: $E[X^k] = \int_{-\infty}^{\infty} x^k f(x)dx$.
- **k-th moment about mean**: $E[(X-\mu)^k] = \int_{-\infty}^{\infty} (x-\mu)^k f(x)dx$.
- **Expected value of squared**: $E[X^2] = \int_{-\infty}^{\infty} x^2 f(x)dx$.

Remark

In general, $E[g(X)] \neq g(E[X])$ (Jensen's Inequality for non-linear g).
Example: $E[X^2] \neq (E[X])^2$ in general.

Examples

Example 11: Uniform Distribution on $[a, b]$

PDF: $f(x) = \begin{cases} \frac{1}{b-a} & \text{if } a \leq x \leq b \\ 0 & \text{otherwise.} \end{cases}$

Find $E[X]$.

Example 12: Exponential distribution with parameter λ

PDF: $f(x) = \begin{cases} \lambda e^{-\lambda x} & \text{if } x \geq 0 \\ 0 & \text{if } x < 0. \end{cases}$

Find $E[X]$.

Variance

Definition 4

The variance *of a continuous random variable X is:*

$$\sigma^2 = Var(X) = E[(X-\mu)^2] = \int_{-\infty}^{\infty} (x-\mu)^2 f(x)dx.$$

Computational Formula

A more convenient formula for calculation:

$$\text{Var}(X) = E[X^2] - (E[X])^2 = E[X^2] - \mu^2.$$

Standard Deviation

$$\sigma = \sqrt{\text{Var}(X)}.$$

Standard deviation has the same units as X.

Interpretation of Variance and Standard Deviation

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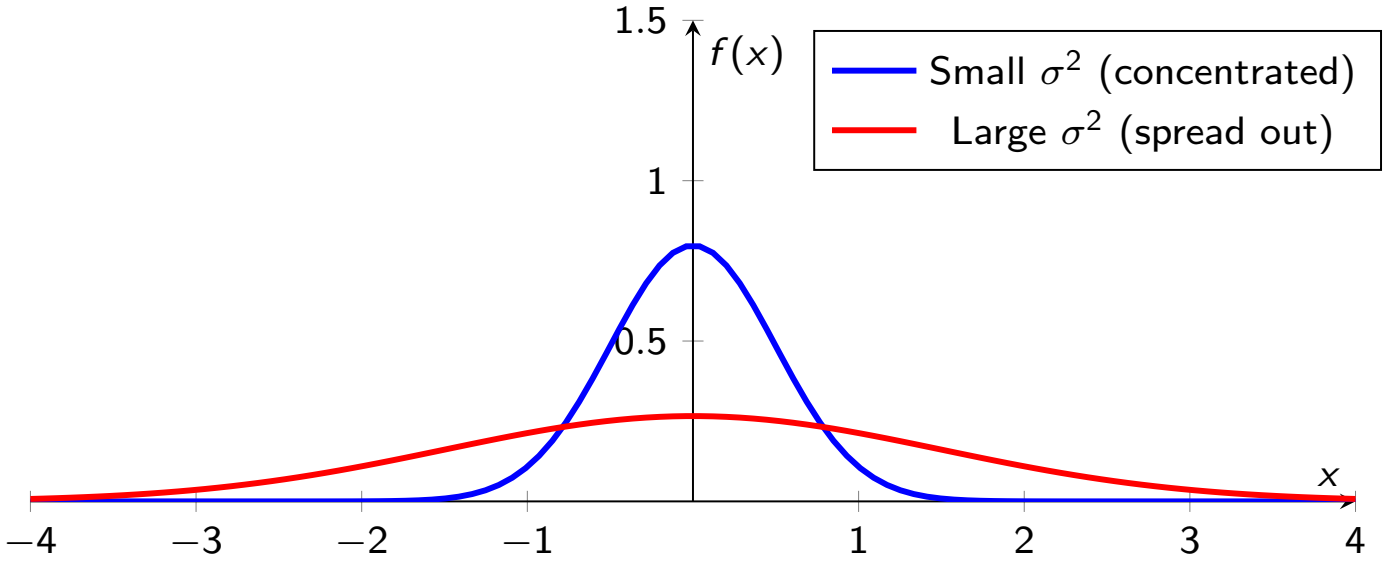
What Does Variance Measure?

- **Spread** or **dispersion** around the mean
- **Variability** or **uncertainty** in the distribution
- Average squared distance from the mean
- Higher variance = more spread out distribution
- Lower variance = more concentrated around mean



Properties of Variance

Proposition 2

For constants a, b, c and random variable X:

1. $Var(X) \geq 0$ *(variance is always non-negative);*
2. $Var(c) = 0$;
3. $Var(aX + b) = a^2 Var(X)$ *(constant shifts don't affect variance);*
4. $Var(aX) = a^2 Var(X)$;
5. $Var(X + b) = Var(X)$ *(adding a constant doesn't change variance);*

Important Note

Unlike expectation, variance is **NOT** linear:

$$\mathrm{Var}(X + Y) \neq \mathrm{Var}(X) + \mathrm{Var}(Y) \text{ in general.}$$

This only holds when X and Y are **independent**:

$$\mathrm{Var}(X + Y) = \mathrm{Var}(X) + \mathrm{Var}(Y).$$

Examples

Example 13: Uniform Distribution on $[a, b]$

PDF: $f(x) = \begin{cases} \frac{1}{b-a} & \text{if } a \leq x \leq b \\ 0 & \text{otherwise.} \end{cases}$

Find $\text{Var}(X)$.

Example 14: Exponential distribution with parameter λ

PDF: $f(x) = \begin{cases} \lambda e^{-\lambda x} & \text{if } x \geq 0 \\ 0 & \text{if } x < 0. \end{cases}$

Find $\text{Var}(X)$.

Examples

Example 15

Let X be a RV with PDF: $f(x) = \begin{cases} Cx^2 & \text{if } 0 \leq x \leq 1 \\ 0 & \text{otherwise.} \end{cases}$

a. Find C.
b. Find $E[X]$.
c. Find $E[X^2]$ then deduce $\text{Var}(X)$ and σ.

Median

Definition 5

The median *m of a continuous random variable X is the value that satisfies:*

$$P(X \leq m) = P(X \geq m) = \frac{1}{2}.$$

Or equivalently, using the CDF:

$$F(m) = \frac{1}{2} \quad \text{or} \quad \int_{-\infty}^{m} f(x)dx = \frac{1}{2}.$$

Properties

- The median divides the distribution into two equal halves;
- It is the 50th percentile;
- More robust to outliers than the mean;
- For symmetric distributions: median $=$ mean;
- For skewed distributions: median $\neq$ mean.

Example: Median of Exponential Distribution

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Example 16: Exponential Distribution with $\lambda = 1$

Let X be a RV with CDF:

$$F(x) = 1 - e^{-x} \text{ for } x \geq 0.$$

Find the median m of X.

Solution:

$$\begin{aligned} F(m) &= 0.5 \\ 1 - e^{-m} &= 0.5 \\ e^{-m} &= 0.5 \\ -m &= \ln(0.5) = -\ln(2) \\ m &= \ln(2) \approx 0.693 \end{aligned}$$

Comparison:

- Mean: $\mu = \frac{1}{\lambda} = 1$.
- Median: $m = \ln(2) \approx 0.693$.
- Median $<$ Mean (right-skewed distribution).

Mode

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Definition 6

The mode *of a continuous random variable X is the value where the PDF is maximized*

$$Mode = \arg\max_x f(x).$$

Mathematically, find x^* such that

$$f'(x^*) = 0 \quad \textit{and} \quad f''(x^*) < 0.$$

Properties

- Represents the "most likely" value (highest density)
- A distribution can have:
  - No mode (uniform distribution)
  - One mode (unimodal)
  - Multiple modes (bimodal, multimodal)
- Less commonly used than mean and median

Uniform Distribution

Definition 7

A random variable X has a uniform distribution *on $[a, b]$, denoted $X \sim Uniform(a, b)$ or $X \sim U(a, b)$, if it has PDF*

$$f(x) = \begin{cases} \frac{1}{b-a} & \text{if } a \leq x \leq b \\ 0 & \text{otherwise.} \end{cases}$$

The CDF of X is given by

$$F(x) = \begin{cases} 0 & \text{if } x < a \\ \frac{x-a}{b-a} & \text{if } a \leq x \leq b \\ 1 & \text{if } x > b. \end{cases}$$

Properties:

- Mean: $\mu = \frac{a+b}{2}$
- Variance: $\sigma^2 = \frac{(b-a)^2}{12}$
- All values equally likely

Applications:

- Random number generation
- Rounding errors
- Arrival times (unknown)
- Initial assumptions

Exponential Distribution: Introduction

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Definition 8

A random variable X has an exponential distribution with rate parameter $\lambda > 0$, denoted $X \sim Exp(\lambda)$, if it has PDF

$$f(x) = \begin{cases} \lambda e^{-\lambda x} & \text{if } x \geq 0 \\ 0 & \text{if } x < 0. \end{cases}$$

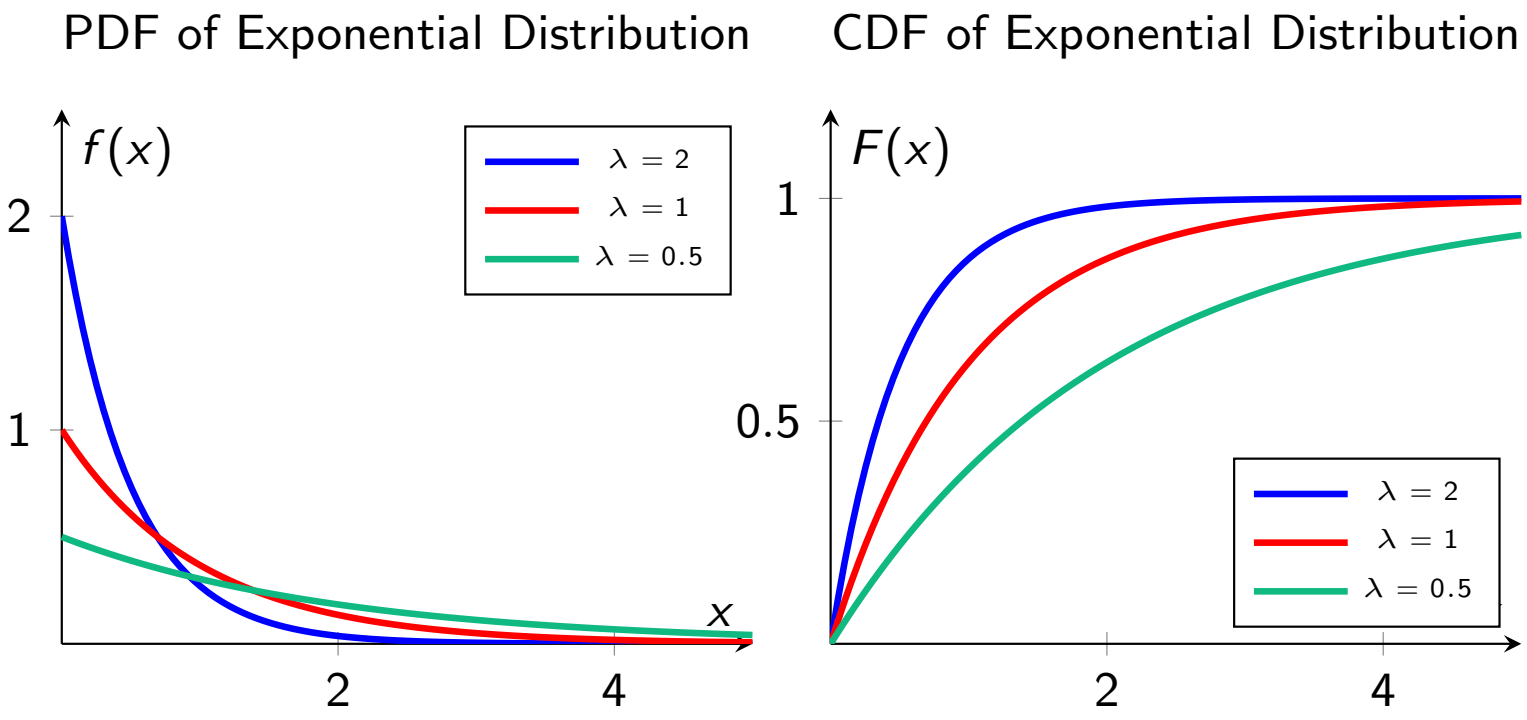
The corresponding CDF is

$$F(x) = \begin{cases} 1 - e^{-\lambda x} & \text{if } x \geq 0 \\ 0 & \text{if } x < 0. \end{cases}$$

Interpretation

- $\lambda =$ rate of occurrence (events per unit time).
- $1/\lambda =$ average waiting time between events.
- Models time until an event occurs.

Exponential Distribution: PDF and CDF



Observation: Larger $\lambda \to$ steeper decay $\to$ shorter waiting times.

Exponential Distribution: Properties

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Gaussian

Basic Properties

For $X \sim \text{Exp}(\lambda)$:

- **Mean (Expected Value):** $E[X] = \frac{1}{\lambda}$.
- **Variance:** $\text{Var}(X) = \frac{1}{\lambda^2}$.
- **Standard Deviation:** $\sigma = \frac{1}{\lambda}$.
- **Median:** $m = \frac{\ln 2}{\lambda} \approx \frac{0.693}{\lambda}$.
- **Mode:** 0 (PDF is highest at $x = 0$).

Useful Formulas

- $P(X > t) = e^{-\lambda t}$ (survival function).
- $P(X \leq t) = 1 - e^{-\lambda t}$.
- $P(a < X < b) = e^{-\lambda a} - e^{-\lambda b}$.

The Memoryless Property

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Memoryless Property

The exponential distribution has a unique **memoryless property**:

$$P(X > s + t \mid X > s) = P(X > t) \quad \forall s, t \geq 0$$

Interpretation: If you've already waited s units without the event occurring, the probability of waiting an additional t units is the same as if you just started waiting.

Example: Light Bulb

A light bulb with exponential lifetime:

- Already used for 100 hours without failure
- Probability of lasting another 50 hours = same as a new bulb lasting 50 hours
- "The bulb doesn't remember it's been used"

Important

The exponential distribution is the **only continuous distribution** with this property!

Applications of Exponential Distribution

Waiting Times:

- Time until next customer arrives
- Time until next phone call
- Time between earthquakes
- Time until radioactive decay

Lifetimes:

- Electronic component failure
- Machine breakdown
- Time to system failure

Queuing Theory:

- Service times
- Inter-arrival times
- Call center operations

Other Applications:

- Distance between events
- Financial modeling
- Reliability analysis
- Survival analysis

Connection to Poisson Process

If events occur according to a Poisson process with rate λ, then the time between consecutive events follows $\text{Exp}(\lambda)$.

Examples

Example 17: Customer Service

The time (in minutes) between customer arrivals at a store follows an exponential distribution with mean 5 minutes.

1. What is the rate parameter λ?
2. What is the probability that the next customer arrives within 3 minutes?
3. Given that no customer has arrived in the first 4 minutes, what is the probability that you wait at least 2 more minutes?

Example 18: Equipment Failure

A machine's lifetime follows an exponential distribution with $\lambda = 0.1$ per year.

1. Find the probability the machine lasts more than 10 years
2. Find the median lifetime
3. Find the 90th percentile

Normal Distribution: Introduction

Definition 9

A random variable X has a normal (Gaussian) distribution *with parameters μ and σ^2, denoted $X \sim \mathcal{N}(\mu, \sigma^2)$, if the PDF of X is*

$$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right), \quad -\infty < x < \infty.$$

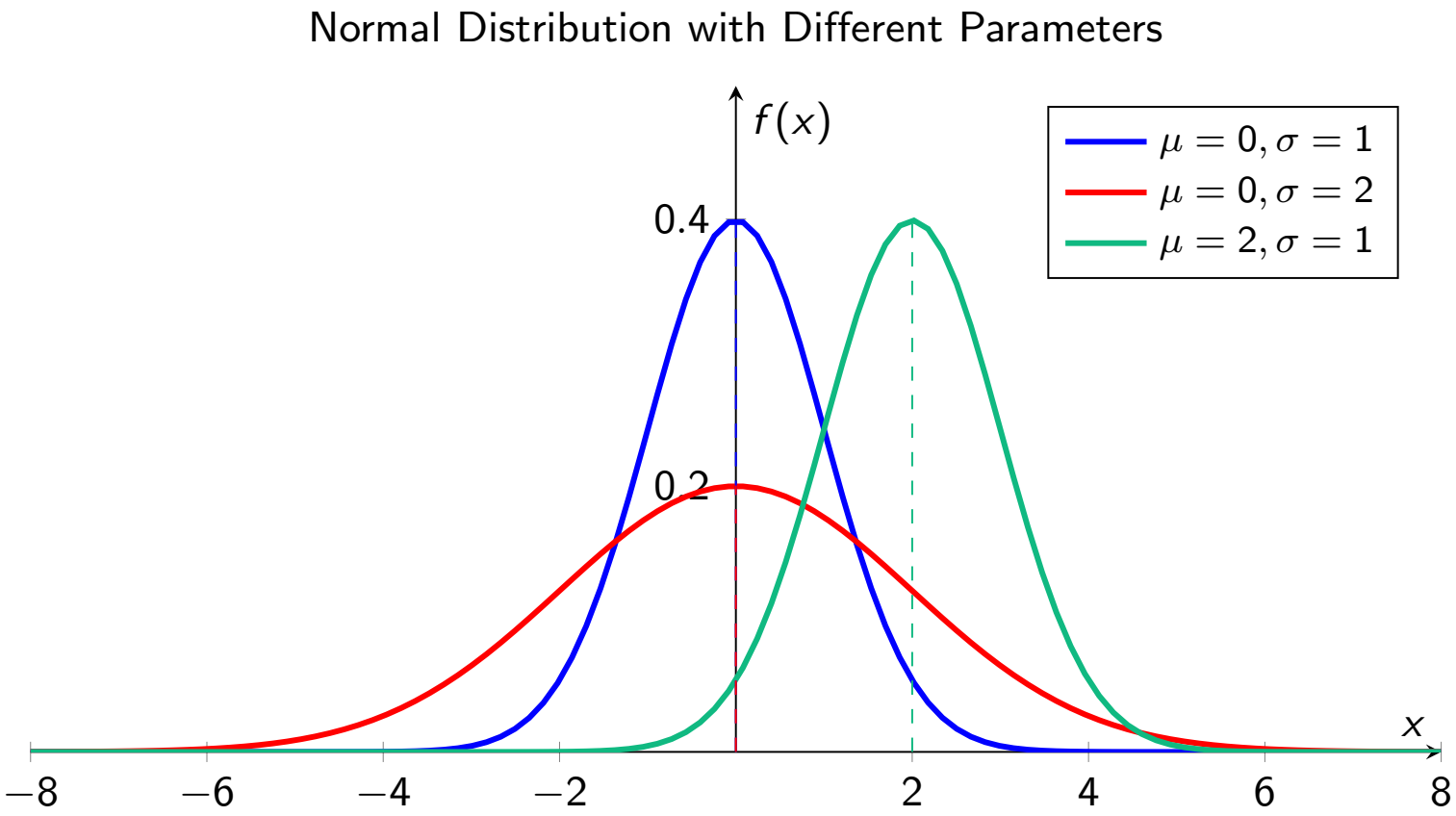
Parameters

- μ = mean (location parameter): center of distribution.
- σ^2 = variance (scale parameter): spread of distribution.
- σ = standard deviation.

Important Notes

- No closed-form expression for CDF.
- CDF is denoted $\Phi(x)$ for standard normal.
- Use tables or software to compute probabilities.

Normal Distribution: PDF Shape



- Bell-shaped, symmetric about μ.
- Peak at $x = \mu$ (mean = median = mode).
- Larger $\sigma \rightarrow$ flatter and wider curve.

Normal Distribution: Properties

Basic Properties

For $X \sim \mathcal{N}(\mu, \sigma^2)$:

- **Mean:** $E[X] = \mu$.
- **Variance:** $\text{Var}(X) = \sigma^2$.
- **Median:** μ (due to symmetry).
- **Mode:** μ (PDF maximum).
- **Symmetry:** $f(\mu + x) = f(\mu - x)$.

Shape Characteristics

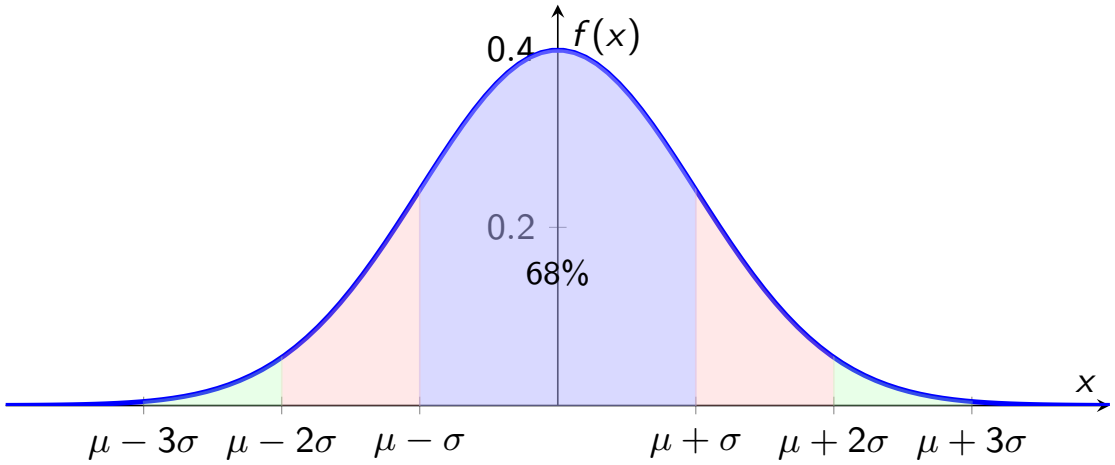
- **Inflection points:** at $x = \mu \pm \sigma$.
- **Support:** $(-\infty, \infty)$ (entire real line).
- **Tails:** Extend to infinity but approach zero quickly.

The Empirical Rule (68-95-99.7 Rule)

Empirical Rule

For $X \sim \mathcal{N}(\mu, \sigma^2)$:

- $P(\mu - \sigma < X < \mu + \sigma) \approx 0.68$ (68%)
- $P(\mu - 2\sigma < X < \mu + 2\sigma) \approx 0.95$ (95%)
- $P(\mu - 3\sigma < X < \mu + 3\sigma) \approx 0.997$ (99.7%)



Standard Normal Distribution

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Definition 10

The standard normal distribution *is* $Z \sim \mathcal{N}(0,1)$ *with PDF* $\phi(z)$ *and CDF* $\Phi(z)$ *given by*

$$\phi(z) = \frac{1}{\sqrt{2\pi}} e^{-z^2/2},$$

$$\Phi(z) = P(Z \leq z) = \int_{-\infty}^{z} \phi(t)dt.$$

Standardization

Any normal random variable can be standardized

$$\text{If } X \sim \mathcal{N}(\mu, \sigma^2), \text{ then } Z = \frac{X-\mu}{\sigma} \sim \mathcal{N}(0,1)$$

This allows us to use **standard normal tables** to compute probabilities.

Key Property

$$P(X \leq x) = P\left(\frac{X-\mu}{\sigma} \leq \frac{x-\mu}{\sigma}\right) = \Phi\left(\frac{x-\mu}{\sigma}\right).$$

Computing Probabilities Using Z-table

Steps to Compute $P(a < X < b)$ for $X \sim \mathcal{N}(\mu, \sigma^2)$

Step 1: Standardize the values

$$z_a = \frac{a - \mu}{\sigma}, \quad z_b = \frac{b - \mu}{\sigma}.$$

Step 2: Use standard normal table

$$P(a < X < b) = P(z_a < Z < z_b) = \Phi(z_b) - \Phi(z_a).$$

Step 3: Look up $\Phi(z_a)$ and $\Phi(z_b)$ in the Z-table.

Common Z-values to Remember

- $\Phi(0) = 0.5$ (by symmetry).
- $\Phi(1) \approx 0.8413 \rightarrow P(Z < 1) = 84.13\%$.
- $\Phi(1.96) \approx 0.975 \rightarrow P(Z < 1.96) = 97.5\%$.
- $\Phi(2) \approx 0.9772 \rightarrow P(Z < 2) = 97.72\%$.
- $P(-1.96 < Z < 1.96) = 0.95$ (important for confidence intervals).

Examples

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Example 19: IQ Scores

IQ scores are normally distributed with mean 100 and standard deviation 15.

1. What proportion of the population has IQ above 130?
2. What proportion has IQ between 85 and 115?
3. Find the 90th percentile of IQ scores

Example 20: Manufacturing Quality Control

A factory produces bolts with lengths that are normally distributed with mean 10 cm and standard deviation 0.2 cm. Bolts are acceptable if their length is between 9.7 cm and 10.3 cm.

1. What proportion of bolts are acceptable?
2. If the factory produces 10,000 bolts per day, how many are expected to be rejected?

Linear Combinations of Normal Variables

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Theorem 2

If $X \sim \mathcal{N}(\mu, \sigma^2)$, then for constants a and b

$$Y = aX + b \sim \mathcal{N}(a\mu + b, a^2\sigma^2).$$

Multiple Independent Normal Variables

If $X_1 \sim \mathcal{N}(\mu_1, \sigma_1^2)$ and $X_2 \sim \mathcal{N}(\mu_2, \sigma_2^2)$ are independent, then:

$$X_1 + X_2 \sim \mathcal{N}(\mu_1 + \mu_2, \sigma_1^2 + \sigma_2^2).$$

More generally, for independent $X_i \sim \mathcal{N}(\mu_i, \sigma_i^2)$:

$$\sum_{i=1}^{n} a_i X_i \sim \mathcal{N}\left(\sum_{i=1}^{n} a_i \mu_i, \sum_{i=1}^{n} a_i^2 \sigma_i^2\right).$$

Applications of Normal Distribution

Natural Phenomena:

- Human heights, weights
- Blood pressure, IQ scores
- Measurement errors
- Test scores
- Particle velocities

Quality Control:

- Product dimensions
- Manufacturing tolerances
- Process variation

Finance:

- Stock returns
- Portfolio analysis
- Risk management
- Black-Scholes model

Statistical Inference:

- Sampling distributions
- Confidence intervals
- Hypothesis testing
- Regression analysis

Central Limit Theorem

The sum (or average) of many independent random variables tends to be normally distributed, regardless of their individual distributions!